



Module Introduction Transcript

Hi everyone, the topic that we are going to study now is the heart of statistics, actually the heart of the subject that we are learning and it deals with a lot of interesting things that we do in statistics. But before we begin and delve deeper into all the content that we have to cover, let me give you a brief idea about what exactly this module is about, so a high level idea of what we are going to study, what we are going to get introduced to and so on. So first thing is, as you can see, it's all about inferential statistics.

And what do you mean by inferential statistics, there are two parts to it, inference and statistics. Inference, as you know, the dictionary meaning is making or learning to know about anything. Statistics on the other hand is the process of sampling and then inferring about whatever you're sampling from.

So inferential statistics is to learn via doing a sample. That's what it is. And in this entire module, we will be going through how do we sample, what different types of samples are there, what different types of estimates do we deduce from the sample.

And then how do we infer about the population using the sample, all of those things would be introduced. And we will be getting deeper into the details of each and every topic layer by layer. So with that, let me start with it.

But let me also switch to board and help you understand the idea of this entire thing. As I was telling you about this entire module is the heart of what we are going to learn in the subject. So just to give you an introduction about what exactly is inferential statistics, let me give you a very, very small example.

I mean, a very convenient one as well, because let's say you have to estimate an average height of people living in India. India is the population that we have and the parameter that we want to estimate is height. This is the parameter.

We generally call it population parameter. Now what are the methods that we have for estimating average height of people living in India? There are two different things. One, you estimate every each and every individual of the population.

So all 140 billion people, that's one of the ways or you take small samples. Now the problem with taking to investigate the entire population is there are three different problems. One, it will take a lot of time.

Second, it will take a lot of resources and majority of the times it's also not infeasible. It's not feasible to take these things. So on the other hand, what we have is we can take smaller samples and using those samples, we can infer about the population.



How do we do that? So let's say if our ask is to estimate the height of people living in India, we can estimate that by taking smaller samples. So if we have these samples, let's say we take a sample of 50 people, now first individual could have a height of 5.5, second individual could have a height of 5.6 and then third could have a height of 4 and then so on. Similarly you could have height of 100 people, the last one could have a height of 6. Now if you think of, we can take average of all of them, it very well can land up to 5.5. But this is an estimate, this value is an estimate.

We discussed two things, one, we would want to know about population's parameter, which is the height, we have landed using the sample 5.5, which is the estimate of population parameter. It may or may not be close to the population parameter. So we have discussed or discovered two things, one, what is a parameter and what is its estimate.

So this entire statistics or influential statistics would revolve around how do we estimate the population parameter using different types of methods that we have and we will investigate or get into the details of different types of inferential statistical methods. And also before that I introduced you to a concept of sampling. How do you, how would you measure even what type of sample is a good representation of a population, whether a small sample, do I have to take 50 people, 50 odd people to estimate about average height of people living in India or do I have to take 500 people to estimate about the same, right, even 5 lakh people and so on.

So how would you say what is the exact sample size and also what are different types of sampling methods that we have. So all of these things is what we are going to learn and I will give you a quick introduction starting from samples, sampling procedures, then we will get into inferential statistics within which we'll start talking about point estimates, then we will move into interval estimates and then so on. So these are the topics that we'll be covering in this module.

I hope you will enjoy all of those.

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